

***B.Tech. Degree III Semester Special Supplementary
Examination June 2013***

**CS 302 LOGIC DESIGN
(2006 Scheme)**

Time : 3 Hours

Maximum Marks : 100

**PART A
(Answer ALL questions)**

(8 × 5 = 40)

- I. (a) Simplify the expressions $AB + A\bar{C} + A\bar{B}C(AB + C)$ using Boolean laws.
(b) Add the following BCD numbers.
(i) $01100111 + 01010011$
(ii) $00100011 + 00010101$
(c) What is a multiplexer? Implement a 4 : 1 multiplexer using basic gates.
(d) What are universal gates? Why they are called universal gates?
(e) What is race around condition? How it can be avoided?
(f) Design a T flip flop using JK flip flops.
(g) What is tristate logic?
(h) Explain the terms fan in, fan out, noise margin, propagation delay, power dissipation with respect to digital IC's.

PART B

(4 × 15 = 60)

- II. (a) Simplify the expression using K-map and realize using NOR gates (7½)
 $F = \sum m(1, 5, 6, 12, 13, 14) + d(2, 4)$.
(b) (i) $(1100)_{\text{gray}} = (A1)_2$ (7½)
(ii) $(141737)_8 = (A2)_{16}$
(iii) $(67.721)_{10} = (A3)_2$
(iv) $(10110110)_2 = (A4)_8$
(v) $(4057.06)_8 = (A5)_{10}$
Find A1, A2, A3, A4 and A5.

OR

- III. (a) Minimize the expression and implement it by using only NOR gates. (7½)
 $F(P, Q, R, S) = \pi M(1, 4, 5, 7, 10, 12, 14, 15) \cdot d(3, 6)$.
(b) What is an excess-3 code? Write one of its key feature. (7½)

(P.T.O.)

- IV. (a) Implement a full adder circuit using NAND gates only. (5)
- (b) Use a multiplexer to implement the function $F = A \oplus B \oplus C$ (5)
- (c) Describe the difference between PLA and PAL. (5)
- OR**
- V. (a) Explain how a look ahead adder speeds up the addition process. (10)
- (b) Expand $A + B\bar{C} + AB\bar{D} + ABCD$ to min terms and max terms. (5)
- VI. (a) Design a mod 6 asynchronous up counter using T flip flops. (6)
- (b) Differentiate between combinational and sequential circuits. (4)
- (c) Draw the circuit diagram and waveform of a 4 bit SISO shift register. (5)
- OR**
- VII. (a) Draw and explain the working of a 4 bit Johnson counter. Draw the timing diagram also. (7½)
- (b) Design a BCD counter using JK flip flops. (7½)
- VIII. (a) Explain the operation of a standard 2 input TTL NAND gate with circuit diagram. (10)
- (b) Explain how TTL to CMOS interface can be done. (5)
- OR**
- IX. (a) Compare ECL and TTL families. (5)
- (b) Explain the operation of a 2 input CMOS NOR gate and CMOS inverter in detail. (10)
